

Board of Studies (BoS) Syllabus Template

Course Information

Degree: UG

Program: BCA

Course Name: OPERATING SYSTEM

Course Credits: 3

Semester: 2

Course Category: Core

Type of Course: Lecture

New Course: No

Learning Hours

Activity	Hours per Week
Lecture	3
Tutorial	0
Practical	0
Experiential Learning	0

Course Contribution

Category	Percentage Contribution
Employment Enhancement	20%
Entrepreneurship Skills	20%
Skill Development	60%

Course Relevance

Level	Percentage Relevance
Global	30%
National	30%
Regional	20%
Local	20%

Course Outcomes (COs)

CO#	Course Outcome	Bloom's Taxonomy Level
CO1	Explain the fundamental components of a computer operating system.	BL-2
CO2	Discuss the concept of process management, threading and analyze different scheduling algorithms.	BL-5
CO3	Solve process synchronization problems using different methods.	BL-3
CO4	Demonstrate the memory and storage management schemes.	BL-4
CO5	Illustrate the implementation of file system and disk management.	BL-3

Module	Course Content	Tools Used / Assessment and Activity	Bloom's Level	Mapping COs	Percentage change in Module (put Nil if there is no change or it is a new course write NEW)
Module 1: Introduction to Operating System	Introduction, Functions of OS	PPT Presentation, MCQs / Open Discussion	BL-2	1	Nil
	Evolution of OS, OS Structures		BL-2	1	Nil
	OS Services, System calls, Types of System calls		BL-2	1	Nil
	System programs, Virtual Machines		BL3	1	Nil
	Mobile Operating System: Introduction Types of MOS		BL-3	1	Nil
	Functions of MOS - Architecture of MOS		BL-2	1	Nil
	Various Operating Systems and its Benefits – Case Study		BL-4	1	NEW
Module 2: Process Management	Process Management	PPT Presentation and Classroom Group Interaction	BL-2	1	Nil
	Processes: Process concept		BL-2	1	Nil
	Operations on processes		BL-2	2	Nil
	Inter process communication		BL-2	2	Nil
	Communication in client-server: Socket		BL-2	2	Nil

	RPC & RMI.	Activity 1: Coursera Certification	BL-2	2	Nil
	Threads: Introduction to Threads, Single and Multithreaded processes and its benefits		BL-2	2	Nil
	User and Kernel threads, Multithreading model		BL-2	2	Nil
	Threading issues		BL-4	2	Nil
	Process Scheduling: Basic concepts		BL-3	2	Nil
	Scheduling criteria, Scheduling Algorithms		BL-4	3	Nil
Module 3: Process Synchronization and Deadlocks	Process Synchronization	Case Study and Group Presentation on Deadlocks	BL-2	3	Nil
	Deadlocks & Mutual Exclusion		BL-2	3	Nil
	Critical – section problem		BL-2	4	Nil
	Peterson's solution		BL-3	4	Nil
	Semaphores		BL-2	3	Nil
	Classic problems of synchronization		BL-2	3	Nil
	Deadlocks: System Model		BL-2	3	Nil
	Deadlock characterization		BL-2	3	Nil
	Methods for handling Deadlock and		BL-4	3	Nil
	Deadlock prevention		BL-3	4	Nil
	Deadlock Avoidance, Detection and Recovery		BL-3	4	Nil
Module 4: Memory and Storage Management	Memory management- logical and physical addresses, Contiguous memory allocation	PPT Presentation And Learn and Present on Virtual Machine	BL-2	4	Nil
	Paging, Segmentation with Paging, Structure of page table		BL-3	4	Nil
	Swapping - Virtual Memory - Demand paging,		BL-3	5	Nil
	Copy-on-write and page replacement Algorithms.		BL-3	5	Nil
Module 5: File-System and Security	File-concept, Access Methods,	PPT Presentation	BL-2	5	Nil
	File System structure, File-System Implementations, Directory Implementation,		BL-2	5	Nil
	Allocation Methods, Free-Space Management, Recovery		BL-4	5	Nil

	Disk Management: Disk Structure, Disk Attachment, Disk Scheduling and Algorithms,		BL-5	4	Nil
	Disk Management, Swap-Space management		BL-4	5	Nil
	Protection: Goals of Protection, Domain of Protection, Access Matrix, Implementation of Access Matrix		BL-4	5	Nil

Text Books (TB):

TB-1	Operating System Concepts, Tanenbaum, 5th Edition, Pearson Education
TB-2	Operating System Concepts, 10th edition Peter B. Galvin, Greg Gagne, Abraham Silberschatz, John Wiley & Sons, Inc

Reference Books (RB):

RB-1	Andrew Tanenbaum, Modern Operating Systems, Prentice Hall.
RB-2	Andrew Tanenbaum & Albert Woodhull, Operating Systems: Design and Implementation. Prentice-Hall.
RB-3	William Stallings, Operating Systems, Prentice Hall.

Other Reading (OR):

OR-1	Operating system fundamentals https://www.tutorialspoint.com/operating_system/index.htm
OR-2	Inter Process Communication (IPC) https://www.geeksforgeeks.org/inter-process-communication-ipc/?ref=leftbar-rightbar

OR-3	CPU Scheduling https://www.javatpoint.com/os-cpu-scheduling
OR-4	Deadlocks in operating system https://www.guru99.com/deadlock-in-operating-system.html

CO-PO/PSO Mapping

C0	P01	P02	P03	P04	P05	P06	P07	P08	P09
C01	3	2	-	-	-	2	2	-	-
C02	3	3	3	3	3	3	1	2	-
C03	3	3	2	2	-	-	-	3	-
C04	3	2	2	2	2	1	-	-	-
C05	3	2	3	3	2	1	1	2	1